

Namespace Teradyne.FA.DIA.DAS.Adaptive

Classes

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Builds and serializes AdaptiveRequests to JSON for transmission.

[AdaptiveCommunication](#)

This class manages RabbitMQ communication to send adaptive request to the tester host.

[AdaptiveRequest](#)

Represents an adaptive request sent to the ATE system, including test step actions, enable word changes, and target information.

[EnableWordAction](#)

Represents an action applied to a set of enable words for adaptive behavior.

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Defines numerical limits and metadata associated with a test step or limit configuration.

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Represents an action that can be applied to one or more test steps in an adaptive test scenario.

Enums

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Enum specifying the target component for an adaptive request.

[EnableWordActionType](#)

Specifies the type of action to apply on enable words in adaptive control.

[TestStepActionType](#)

Defines the types of actions that can be taken on test steps in an adaptive scenario.

Class AdaptiveCommand

Namespace: [Teradyne.FA.DIA.DAS.Adaptive](#)

Assembly: Adaptive.dll

Builds and serializes AdaptiveRequests to JSON for transmission.

```
public class AdaptiveCommand
```

Inheritance

[object](#)  ← AdaptiveCommand

Inherited Members

[object.Equals\(object\)](#) , [object.Equals\(object, object\)](#) , [object.GetHashCode\(\)](#) , [object.GetType\(\)](#) ,
[object.MemberwiseClone\(\)](#) , [object.ReferenceEquals\(object, object\)](#) , [object.ToString\(\)](#) 

Constructors

AdaptiveCommand(string, List<EnableWordAction>, List<TestStepAction>)

Initializes a new AdaptiveCommand object.

```
public AdaptiveCommand(string DasId, List<EnableWordAction> enableWords = null,  
List<TestStepAction> steps = null)
```

Parameters

DasId [string](#) 

DAS identifier string.

enableWords [List](#)  <[EnableWordAction](#)>

Optional list of EnableWordAction objects.

steps [List](#)  <[TestStepAction](#)>

Optional list of TestStepAction objects.

Methods

addAction(EnableWordAction)

Adds an EnableWordAction to the request.

```
public bool addAction(EnableWordAction action)
```

Parameters

action [EnableWordAction](#)

Returns

[bool](#)

addAction(TestStepAction)

Adds a TestStepAction to the request.

```
public bool addAction(TestStepAction action)
```

Parameters

action [TestStepAction](#)

Returns

[bool](#)

createClearAction()

Creates a TestStepAction that clears all existing configurations.

```
public TestStepAction createClearAction()
```

Returns

createEnableWordAction(EnableWordActionType, List<string>)

Creates a new EnableWordAction.

```
public EnableWordAction createEnableWordAction(EnableWordActionType actionType,  
List<string> enableWords)
```

Parameters

actionType [EnableWordActionType](#)

enableWords [List](#) <[string](#)>

Returns

[EnableWordAction](#)

createLimitsAction(TestStepActionType, List<int>, Limits)

Creates a TestStepAction with limits using step numbers.

```
public TestStepAction createLimitsAction(TestStepActionType actionType, List<int> numbers,  
Limits limits)
```

Parameters

actionType [TestStepActionType](#)

numbers [List](#) <[int](#)>

limits [Limits](#)

Returns

[TestStepAction](#)

createLimitsAction(TestStepActionType, List<string>, Limits)

Creates a TestStepAction with limits using step names.

```
public TestStepAction createLimitsAction(TestStepActionType actionType, List<string> names,
Limits limits)
```

Parameters

actionType [TestStepActionType](#)

names [List](#) <[string](#)>

limits [Limits](#)

Returns

[TestStepAction](#)

createTestStepAction(TestStepActionType, List<int>)

Creates a TestStepAction based on test step numbers.

```
public TestStepAction createTestStepAction(TestStepActionType actionType, List<int> numbers)
```

Parameters

actionType [TestStepActionType](#)

numbers [List](#) <[int](#)>

Returns

[TestStepAction](#)

createTestStepAction(TestStepActionType, List<string>)

Creates a TestStepAction based on test step names.

```
public TestStepAction createTestStepAction(TestStepActionType actionType,  
List<string> names)
```

Parameters

actionType [TestStepActionType](#)

names [List](#) <[string](#)>

Returns

[TestStepAction](#)

toJSON()

Converts the AdaptiveRequest to a formatted JSON string.

```
public string toJSON()
```

Returns

[string](#)

A JSON string, or "" in case of serialization error.

Class AdaptiveCommunication

Namespace: [Teradyne.FA.DIA.DAS.Adaptive](#)

Assembly: Adaptive.dll

This class manages RabbitMQ communication to send adaptive request to the tester host.

```
public class AdaptiveCommunication
```

Inheritance

[object](#) ← AdaptiveCommunication

Inherited Members

[object.Equals\(object\)](#), [object.Equals\(object, object\)](#), [object.GetHashCode\(\)](#), [object.GetType\(\)](#), [object.MemberwiseClone\(\)](#), [object.ReferenceEquals\(object, object\)](#), [object.ToString\(\)](#)

Constructors

AdaptiveCommunication(string)

```
public AdaptiveCommunication(string serverIP)
```

Parameters

serverIP [string](#)

Fields

RMQ_CLIENTNAME

```
public const string RMQ_CLIENTNAME = "AMP DAS RabbitMQ Sender"
```

Field Value

[string](#)

RMQ_EXCHANGE

```
public const string RMQ_EXCHANGE = "AMP_DAS_Exchange"
```

Field Value

[string](#)

RMQ_PORT

```
public const int RMQ_PORT = 5672
```

Field Value

[int](#)

RMQ_PWD

```
public const string RMQ_PWD = "TER-ADAPTIVE-P4SSW0RD"
```

Field Value

[string](#)

RMQ_QUEUE

```
public const string RMQ_QUEUE = "AMP_DAS_Queue"
```

Field Value

[string](#)

RMQ_ROUTING


```
public const string RMQ_ROUTING = "AMP_DAS_RoutingKey"
```

Field Value

[string](#)

RMQ_SCHEME

```
public const string RMQ_SCHEME = "amqp://"
```

Field Value

[string](#)

RMQ_USER

```
public const string RMQ_USER = "AMP-adaptive"
```

Field Value

[string](#)

RMQ_VHOST

```
public const string RMQ_VHOST = "TER-AMP-vhost"
```

Field Value

[string](#)

connectionFactory

`protected` ConnectionFactory connectionFactory

Field Value

ConnectionFactory

m_basicProperties

`protected` IBasicProperties m_basicProperties

Field Value

IBasicProperties

m_channel

`protected` IModel m_channel

Field Value

IModel

m_connection

`protected` IConnection m_connection

Field Value

IConnection

m_consumer

```
protected EventingBasicConsumer m_consumer
```

Field Value

EventingBasicConsumer

m_replyQueueName

```
protected string m_replyQueueName
```

Field Value

[string](#)

m_respQueue

```
protected BlockingCollection<string> m_respQueue
```

Field Value

[BlockingCollection](#) <[string](#)>

rmqServerIP

```
protected string rmqServerIP
```

Field Value

[string](#)

Methods

Close()

Close the connection to the RabbitMQ server if it is open.

```
public void Close()
```

Remarks

Attempting to call `SendAction(AdaptiveControlCommandBase)`

`SendAction(AdaptiveControlCommandBase)` after closure will result in undefined behavior.

InitBasicProperties()

```
protected void InitBasicProperties()
```

InitConnection()

```
protected bool InitConnection()
```

Returns

[bool](#)

InitConsumerReceived()

```
protected void InitConsumerReceived()
```

Open()

Open a connection to the RabbitMQ server.

```
public bool Open()
```

Returns

[bool](#)

SendAction(string)

Send an Adaptive Action to the AMP server.

```
public string SendAction(string message)
```

Parameters

message [string](#)

Adaptive Action to send. Must be in properly-formatted JSON to be processed correctly.

Returns

[string](#)

Response from AMP as a string.

Remarks

See documentation for formatting details. Will throw an InvalidOperationException if the connection is not open.

SendAction(AdaptiveCommand)

Send an Adaptive Action to the AMP server.

```
public string SendAction(AdaptiveCommand command)
```

Parameters

command [AdaptiveCommand](#)

Returns

[string](#)

Response from AMP as a string.

Remarks

See documentation for formatting details. Will throw an `InvalidOperationException` if the connection is not open.

SendActionAsync(string)

Asynchronously send an Adaptive Action to the AMP server.

```
public Task<string> SendActionAsync(string message)
```

Parameters

message [string](#)

Adaptive Action to send. Must be in properly-formatted JSON to be processed correctly.

Returns

[Task](#) <[string](#)>

Response from AMP as a string.

Remarks

See documentation for formatting details. Will throw an `InvalidOperationException` if the connection is not open.

SendActionAsync(AdaptiveCommand)

Asynchronously send an Adaptive Action to the AMP server.

```
public Task<string> SendActionAsync(AdaptiveCommand command)
```

Parameters

command [AdaptiveCommand](#)

Returns

[Task](#) <[string](#)>

Response from AMP as a string.

Remarks

See documentation for formatting details. Will throw an `InvalidOperationException` if the connection is not open.

Class AdaptiveRequest

Namespace: [Teradyne.FA.DIA.DAS.Adaptive](#)

Assembly: Adaptive.dll

Represents an adaptive request sent to the ATE system, including test step actions, enable word changes, and target information.

```
public class AdaptiveRequest
```

Inheritance

[object](#)  ← AdaptiveRequest

Inherited Members

[object.Equals\(object\)](#)  , [object.Equals\(object, object\)](#)  , [object.GetHashCode\(\)](#)  , [object.GetType\(\)](#)  , [object.MemberwiseClone\(\)](#)  , [object.ReferenceEquals\(object, object\)](#)  , [object.ToString\(\)](#) 

Constructors

AdaptiveRequest()

Initializes a new instance of the [AdaptiveRequest](#) class and sets the timestamp to the current UTC time.

```
public AdaptiveRequest()
```

Properties

DAS_IDENTIFIER

The identifier of the DAS instance generating the request.

```
[JsonRequired]  
public string DAS_IDENTIFIER { get; set; }
```

Property Value

[string](#)

ENABLEWORD_ACTIONS

A list of enable word actions to perform (e.g., ENABLE, DISABLE).

```
public List<EnableWordAction> ENABLEWORD_ACTIONS { get; set; }
```

Property Value

[List](#) <[EnableWordAction](#)>

TARGET

The target system or software intended to receive the request.

```
[JsonRequired]  
public AdaptiveRequestTarget TARGET { get; set; }
```

Property Value

[AdaptiveRequestTarget](#)

TESTSTEP_ACTIONS

A list of test step actions to perform (e.g., ENABLE, DISABLE, UPDATE_LIMITS).

```
public List<TestStepAction> TESTSTEP_ACTIONS { get; set; }
```

Property Value

[List](#) <[TestStepAction](#)>

TIME_STAMP

The timestamp when the request was created.

```
[JsonRequired]  
public DateTime TIME_STAMP { get; set; }
```

Property Value

[DateTime](#)

Enum AdaptiveRequestTarget

Namespace: [Teradyne.FA.DIA.DAS.Adaptive](#)

Assembly: Adaptive.dll

Enum specifying the target component for an adaptive request.

```
[JsonConverter(typeof(StringEnumConverter))]
```

```
public enum AdaptiveRequestTarget
```

Fields

```
ATE_SOFTWARE = 0
```

Indicates that the adaptive request is intended for the ATE software (test executive).

Class EnableWordAction

Namespace: [Teradyne.FA.DIA.DAS.Adaptive](#)

Assembly: Adaptive.dll

Represents an action applied to a set of enable words for adaptive behavior.

```
public class EnableWordAction
```

Inheritance

[object](#)  ← EnableWordAction

Inherited Members

[object.Equals\(object\)](#) , [object.Equals\(object, object\)](#) , [object.GetHashCode\(\)](#) , [object.GetType\(\)](#) ,
[object.MemberwiseClone\(\)](#) , [object.ReferenceEquals\(object, object\)](#) , [object.ToString\(\)](#) 

Properties

ACTION

The action to perform on the enable words (ENABLE or DISABLE).

```
[JsonRequired]  
public EnableWordActionType? ACTION { get; set; }
```

Property Value

[EnableWordActionType?](#)

ENABLEWORDS

A list of enable word identifiers to be affected by the action.

```
[JsonRequired]  
public List<string> ENABLEWORDS { get; set; }
```

Property Value

[List](#) <[string](#)>

Enum EnableWordActionType

Namespace: [Teradyne.FA.DIA.DAS.Adaptive](#)

Assembly: Adaptive.dll

Specifies the type of action to apply on enable words in adaptive control.

```
[JsonConverter(typeof(StringEnumConverter))]
```

```
public enum EnableWordActionType
```

Fields

DISABLE = 1

Disables a feature or test based on the enable word.

ENABLE = 0

Enables a feature or test based on the enable word.

Class Limits

Namespace: [Teradyne.FA.DIA.DAS.Adaptive](#)

Assembly: Adaptive.dll

Defines numerical limits and metadata associated with a test step or limit configuration.

```
public class Limits
```

Inheritance

[object](#)  ← Limits

Inherited Members

[object.Equals\(object\)](#) , [object.Equals\(object, object\)](#) , [object.GetHashCode\(\)](#) , [object.GetType\(\)](#) ,
[object.MemberwiseClone\(\)](#) , [object.ReferenceEquals\(object, object\)](#) , [object.ToString\(\)](#) 

Properties

HI_LIMIT

Upper limit value (can be null).

```
[JsonRequired]  
public double? HI_LIMIT { get; set; }
```

Property Value

[double](#) ?

HLM_SCALE

Scale factor applied to the upper limit (optional).

```
public int? HLM_SCALE { get; set; }
```

Property Value

[int](#)?

LIMIT_NAMES

Names of the limits involved (e.g., measurement names or parameter labels).

```
public List<string> LIMIT_NAMES { get; set; }
```

Property Value

[List](#) <[string](#)>

LIMIT_NUMBERS

Numeric IDs corresponding to the limits.

```
public List<int> LIMIT_NUMBERS { get; set; }
```

Property Value

[List](#) <[int](#)>

LLM_SCALE

Scale factor applied to the lower limit (optional).

```
public int? LLM_SCALE { get; set; }
```

Property Value

[int](#)?

LO_LIMIT

Lower limit value (can be null).


```
[JsonRequired]  
public double? LO_LIMIT { get; set; }
```

Property Value

[double](#)?

Class TestStepAction

Namespace: [Teradyne.FA.DIA.DAS.Adaptive](#)

Assembly: Adaptive.dll

Represents an action that can be applied to one or more test steps in an adaptive test scenario.

```
public class TestStepAction
```

Inheritance

[object](#)  ← TestStepAction

Inherited Members

[object.Equals\(object\)](#) , [object.Equals\(object, object\)](#) , [object.GetHashCode\(\)](#) , [object.GetType\(\)](#) ,
[object.MemberwiseClone\(\)](#) , [object.ReferenceEquals\(object, object\)](#) , [object.ToString\(\)](#) 

Properties

ACTION

The type of action to apply (e.g., ENABLE, DISABLE).

```
[JsonRequired]  
public TestStepActionType ACTION { get; set; }
```

Property Value

[TestStepActionType](#)

LIMITS

The limit configuration to be used if the action is UPDATE_LIMITS.

```
public Limits LIMITS { get; set; }
```

Property Value

TESTSTEP_NAMES

A list of test step names to which the action applies.

```
public List<string> TESTSTEP_NAMES { get; set; }
```

Property Value

[List](#) <[string](#)>

TESTSTEP_NUMBERS

A list of test step numbers to which the action applies.

```
public List<int> TESTSTEP_NUMBERS { get; set; }
```

Property Value

[List](#) <[int](#)>

Enum TestStepActionType

Namespace: [Teradyne.FA.DIA.DAS.Adaptive](#)

Assembly: Adaptive.dll

Defines the types of actions that can be taken on test steps in an adaptive scenario.

```
[JsonConverter(typeof(StringEnumConverter))]  
public enum TestStepActionType
```

Fields

CLEAR = 3

Clears configuration for a test step.

DISABLE = 1

Disables a test step.

ENABLE = 0

Enables a test step.

UPDATE_LIMITS = 2

Updates the limit values for a test step.